20261022 Quiz

for (MATH301) Modern Algebra I

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Exercise 4.5.13. Prove that a group of order 56 has a normal Sylow p-subgroup for some prime p dividing its order.

Solution. Since $56 = 2^3 \cdot 7$, Sylow's Theorem gives

$$n_7 \mid 8, \qquad n_7 \equiv 1 \pmod 7,$$

so $n_7 = 1$ or 8. If $n_7 = 1$, the Sylow 7-subgroup is normal.

If $n_7 = 8$, the eight distinct subgroups of order 7 intersect pairwise only in the identity and account for $8(7-1) = 48$ elements of order 7. Thus exactly eight elements of G do not have order 7. Every Sylow 2-subgroup has order 8 and contains no element of order 7, so it consists of precisely these eight elements. Hence the Sylow 2-subgroup is unique and therefore normal. □

Exercise 4.5.16. Let $|G| = pqr$, where p, q, r are primes with $p < q < r$. Prove that G has a normal Sylow subgroup for p, q, or r.

Solution. Sylow's Theorem gives $n_r \mid pq$ and $n_r \equiv 1 \pmod r$. Since $p, q < r$, this forces $n_r = 1$ or pq. In the first case the Sylow r-subgroup is normal.

Suppose that $n_r = pq$. These distinct subgroups of order r account for $pq(r-1)$ nonidentity elements, leaving only $pq-1$ nonidentity elements of other orders. On the other hand, $n_q \mid pr$ and $n_q \equiv 1 \pmod q$. If $n_q > 1$, then $n_q \geq r$, since $p < q$; the Sylow q-subgroups would consequently contribute at least $r(q-1)$ distinct elements of order q. But

$$r(q-1) \geq rp > pq - 1,$$

a contradiction. Hence $n_q = 1$, and the Sylow q-subgroup is normal. □

Exercise 4.5.17. Prove that if $|G| = 105$, then G has a normal Sylow 5-subgroup and a normal Sylow 7-subgroup.

Solution. Since $105 = 3 \cdot 5 \cdot 7$, Sylow's Theorem gives

$$n_5 \in \{1, 21\}, \qquad n_7 \in \{1, 15\}.$$

If both were greater than 1, the distinct Sylow 5- and Sylow 7-subgroups would contain $21(5-1) + 15(7-1) = 174$ distinct nonidentity elements, which is impossible. Thus at least one is normal.

Suppose first that a Sylow 7-subgroup P_7 is normal. The quotient G/P_7 has order 15 and hence has a unique Sylow 5-subgroup. Its preimage K in G is a normal subgroup of order 35. In K, the number of Sylow 5-subgroups divides 7 and is congruent to 1 modulo 5, so it equals 1. Conjugation by any element of G preserves K and therefore its unique Sylow 5-subgroup. Thus a Sylow 5-subgroup of G is normal.

Conversely, if a Sylow 5-subgroup P_5 is normal, then G/P_5 has order 21 and a unique Sylow 7-subgroup. Its preimage is a normal subgroup K of order 35; in K, the Sylow 7-subgroup is unique, since its number divides 5 and is 1 modulo 7. The same conjugation argument shows that it is normal in G.

□

Supplementary Exercise. Let G be a group of order 30. Determine whether G is simple. Justify your answer.

Solution. Sylow's Theorem gives

$$n_5 \in \{1, 6\}, \qquad n_3 \in \{1, 10\}.$$

If G were simple, neither Sylow subgroup could be normal, so $n_5 = 6$ and $n_3 = 10$. The distinct Sylow 5-subgroups would then contain $6(5-1) = 24$ elements of order 5, while the distinct Sylow 3-subgroups would contain $10(3-1) = 20$ elements of order 3. Including the identity gives at least $1 + 24 + 20 = 45$ elements, contradicting $|G| = 30$. Hence G is not simple.

□

Exercise 5.1.18. In each of (a) to (e), give an example of a group with the specified properties:

(a) an infinite group in which every element has order 1 or 2;

(b) an infinite group in which every element has finite order but for each positive integer n there is an element of order n;

(c) a group with an element of infinite order and an element of order 2;

(d) a group G such that every finite group is isomorphic to some subgroup of G;

(e) a nontrivial group G such that $G \cong G \times G$.

Solution.

(a) Take $G = \bigoplus_{n \geq 1} C_2$. It is infinite, and every element has finite support and squares to the identity; thus every nonidentity element has order 2.

(b) Take $G = \bigoplus_{n \geq 1} C_n$. Every element has only finitely many nonidentity coordinates, so its order is finite. The generator in the nth factor has order n for each $n \geq 1$.

(c) Take $G = \mathbb{Z} \times C_2$. The element $(1, 1_{C_2})$ has infinite order, while $(0, t)$ has order 2 when t is the nonidentity element of C_2.

(d) Take $G = \bigcup_{n \geq 1} S_n$, where $S_n \leq S_{n+1}$ fixes $n + 1$. Every finite group H embeds into $S_{|H|}$ by its action on itself by left multiplication (Cayley's Theorem), and hence embeds into G.

(e) Take $G = \bigoplus_{n \geq 1} C_2$ again. Splitting the coordinates into odd and even indices gives $G \cong G \times G$. The group is nontrivial.

□

Exercise 5.2.10. Let n and k be positive integers and let A be the free abelian group of rank n (written additively). Prove that A/kA is isomorphic to the direct product of n copies of $\mathbb{Z}/k\mathbb{Z}$, where

$$kA = \{ka \mid a \in A\}.$$

Hint. See Exercise 5.1.14.

Solution. Choose a basis $a_1, \ldots, a_n$ of A. Each $a \in A$ is uniquely of the form $a = m_1a_1 + \cdots + m_na_n$ with $m_i \in \mathbb{Z}$. Define

$$\varphi\colon A \longrightarrow (\mathbb{Z}/k\mathbb{Z})^n,$$
$$m_1a_1 + \cdots + m_na_n \longmapsto ([m_1]_k, \ldots, [m_n]_k).$$

This is a surjective homomorphism. Moreover, $a \in \ker\varphi$ if and only if $k \mid m_i$ for every i, which is equivalent to $a \in kA$. Thus $\ker\varphi = kA$, and the First Isomorphism Theorem gives

$$A/kA \cong (\mathbb{Z}/k\mathbb{Z})^n.$$

□